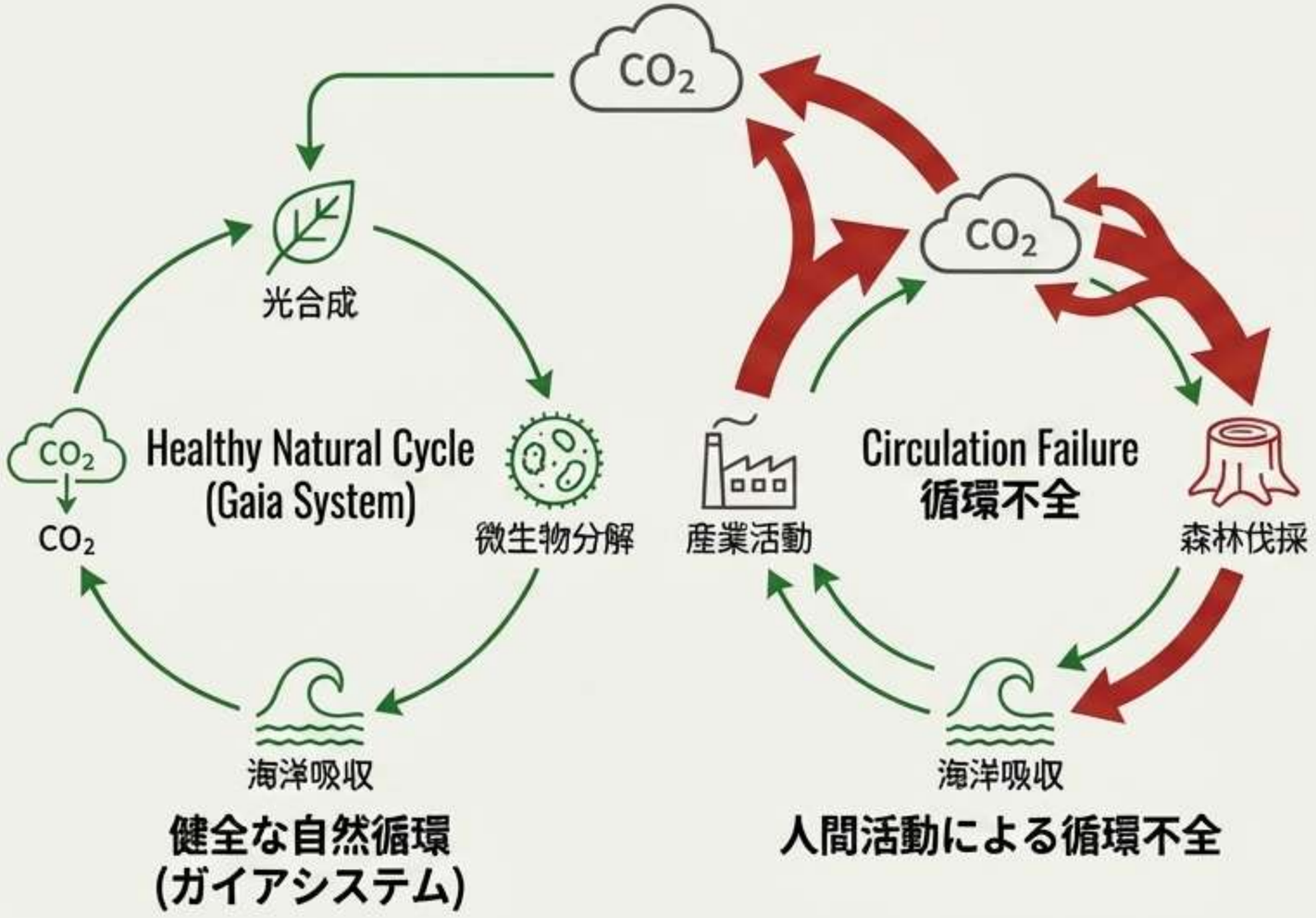

Earth Regeneration Starting from Microorganisms: Blueprint for a New Economic Zone
MBT55 Ecological Hypercycle | AGRIX Digital Trust Layer | Plant-Based Planet Economy
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The Essence of the Crisis We Face: Systemic Circulation Failure of the Earth System



Climate change is not just a CO_2 problem; it is a "breakdown of the carbon cycle" that has exceeded the Earth's self-regulating mechanisms (Gaia System).

- Nature's Regulating Ability: Gaia theory suggests the Earth is a self-regulating system that adjusts greenhouse gas concentrations through photosynthesis, microbial activity, and feedback
- Overload by Human Activity: However, the speed and scale of emissions from modern human activity far exceed this natural regulating ability.
- Required Intervention: To restore the stability of the Gaia system, conscious and scientific intervention to accelerate and optimize the "material cycle," the system's foundation, is essential.

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Catalyst for the Ecological Hypercycle: MBT55

MBT55 is a biological engine that restarts the Earth's self-regulating system. This consortium of over 120 microbial species dramatically accelerates the natural material cycle.

Composition: A precisely controlled microbial community of aerobic (55%) and anaerobic (45%) bacteria. It decomposes all organic matter, including proteins, lipids, cellulose, and lignin.

Core Function: Shortens the organic matter decomposition and fermentation process to 24 hours. This is 90 times faster than conventional microorganisms.

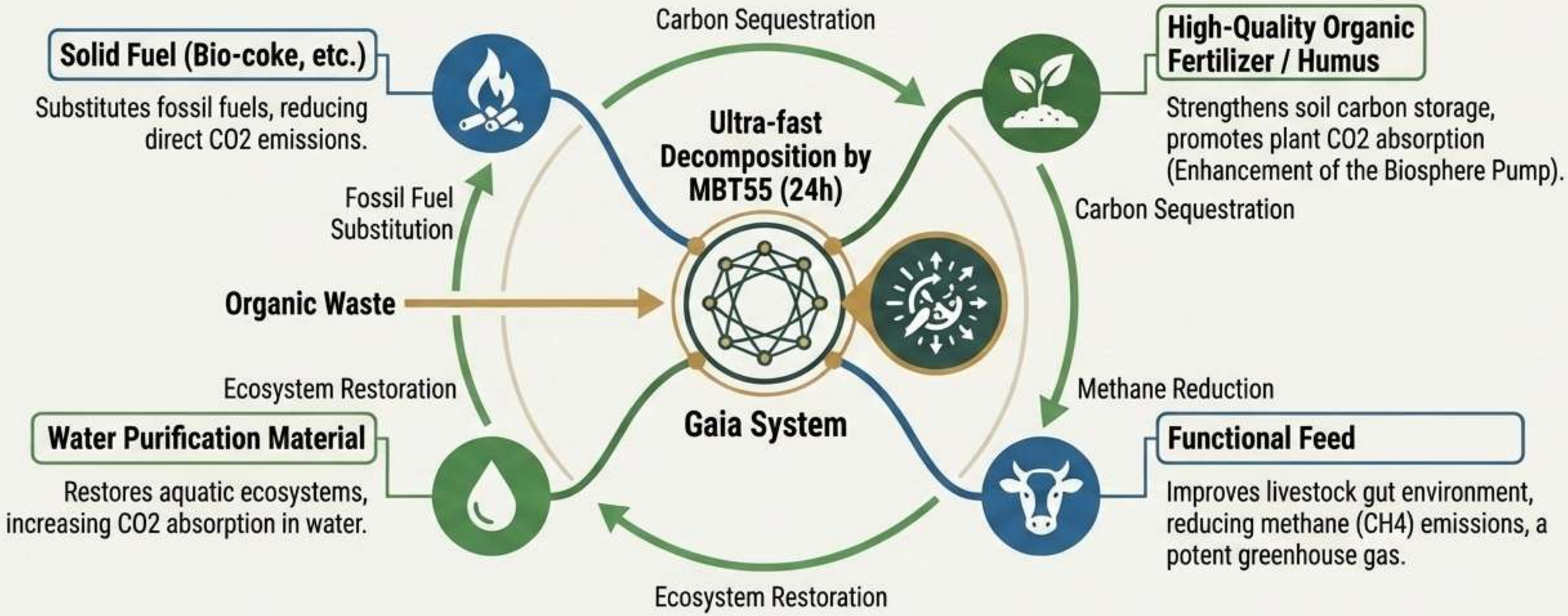
High-Temperature Fermentation: Activates at high temperatures of 80-100°C, converting materials into useful resources while inactivating pathogens and weed seeds.



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Feedback Loop of MBT55 Based on Gaia Theory

MBT55 blocks the pathway for organic waste to become greenhouse gases, building and strengthening a "Resource Recovery → Carbon Sequestration" negative feedback loop. This action allows humanity, as a member of Gaia, to contribute to the system's stability.



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Proof: Regeneration of Tsunami-Damaged Farmland

After the 2011 Great East Japan Earthquake, farmland was covered with seabed sludge containing salt and heavy metals due to the tsunami. MBT55 overturned this hopeless situation in just a few weeks.



Farmland immediately after the disaster (tsunami sludge)

Regeneration Timeline

1st — April 29th
Scattered MBT55 on the sludge.

2nd — 1 Week Later
Activated by natural rainfall after scattering. Sludge decomposition completed, exposing the soil underneath.

3rd — May 20th (21 Days Later)
Rice planting carried out in the paddy field previously deemed unusable.

4th — June 18th (50 Days Later)
Rice grew vigorously in the MBT55 treated plot.





Farmland after MBT55 treatment (regenerated)

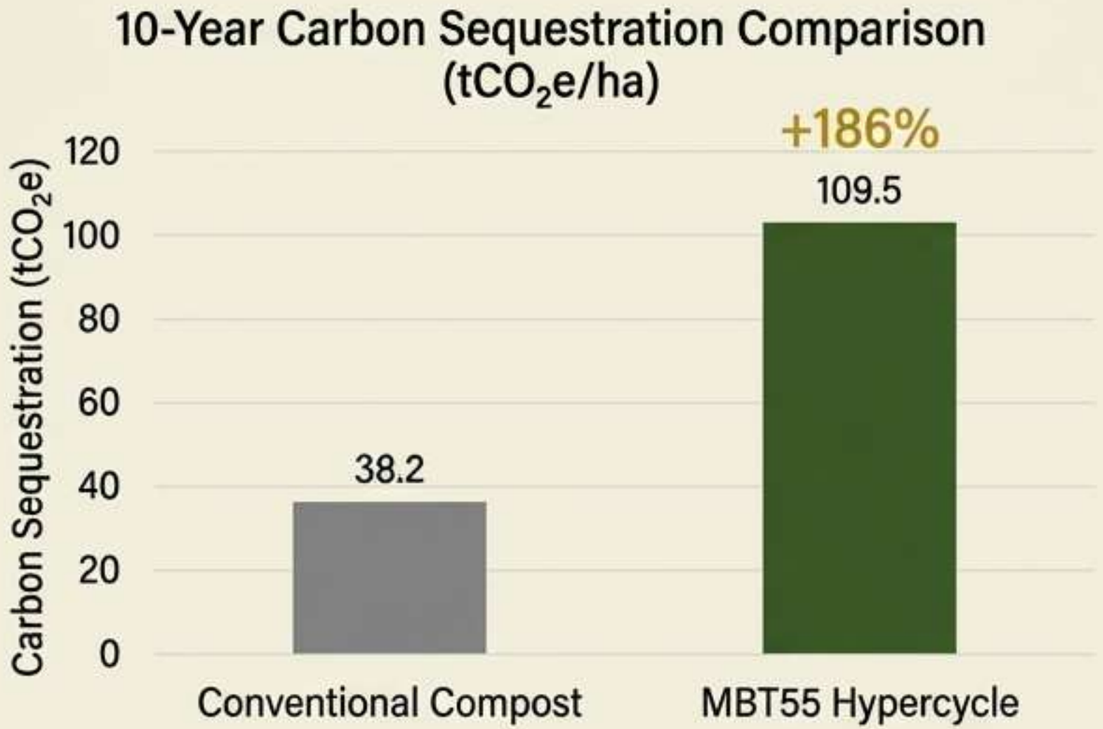
Core Technology

MBT55 converts sludge into organic nutrients while detoxifying heavy metals and chemical pollutants. It activates an ecological hyper-cycle and reconstructs the soil ecosystem at an astonishing speed.

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Quantitative Advantage: Comparison of Carbon Sequestration Capacity

The MBT55 Ecological Hypercycle achieves “cyclic sublimation” of carbon. It is not just simple sequestration, but accelerates the carbon cycle while enhancing ecosystem resilience, and also possesses warming resistance.



10-Year Cumulative Effect Comparison (per hectare)

Indicator	Conventional Compost	MBT Hypercycle	Difference
Carbon Sequestration (tCO_2e)	38.2	109.5	+186%
Humus Production (t)	12.1	35.8	+196%
Methane Emission Reduction Rate (%)	10%	82%	+72% points
Forest Fire Risk Reduction (%)	5%	42%	+37% points

Warming Resistance: When the temperature rises from 35°C to 45°C, the carbon stock of conventional compost decreases by 38%, whereas the MBT55 system only decreases by 8% (4.75 times more resistant).

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Case Study ① Coffee Industry: A Prescription for the Production Crisis (Rust)

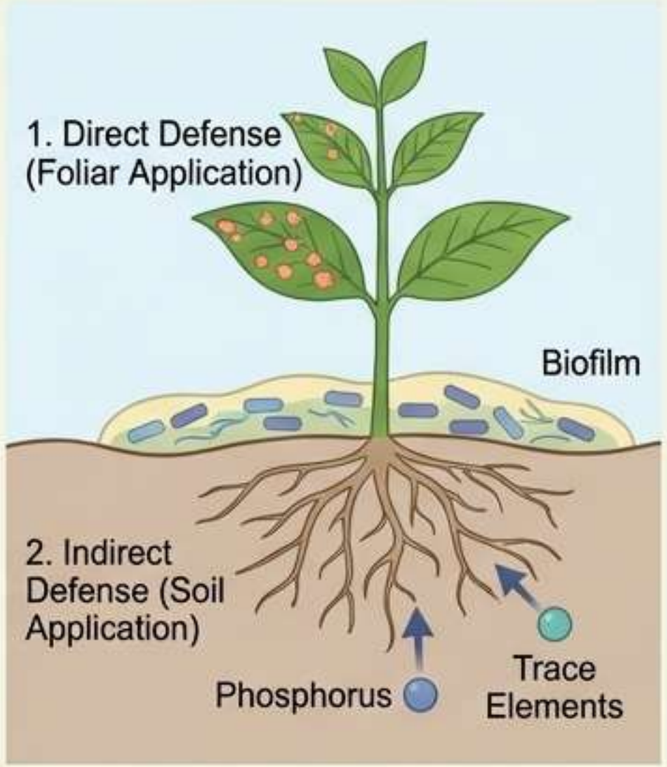
Coffee Leaf Rust threatens to devastate the coffee industry. MBT55 provides a multi-layered, ecosystem-based defense system to replace chemical pesticides.



Coffee leaf immediately after infection (Rust)

【Defense Mechanism by MBT55】

1. Direct Defense (Foliar Application):
   - **Competitive Exclusion**: Beneficial filamentous fungi and actinomycetes form colonies on the leaf surface, physically blocking the entry of rust.
   - **Antimicrobial Substances**: Actinomycetes produce enzymes (like chitinase) that degrade the cell walls of pathogens.
2. Indirect Defense (Soil Application):
   - **Rhizosphere Strengthening**: Reconstructs the rhizosphere microbiota, activating the coffee tree's immunity (Induced Systemic Resistance: ISR).
   - **Nutrient Optimization**: Increases the absorption efficiency of phosphorus and trace elements, improving the tree's stress tolerance.

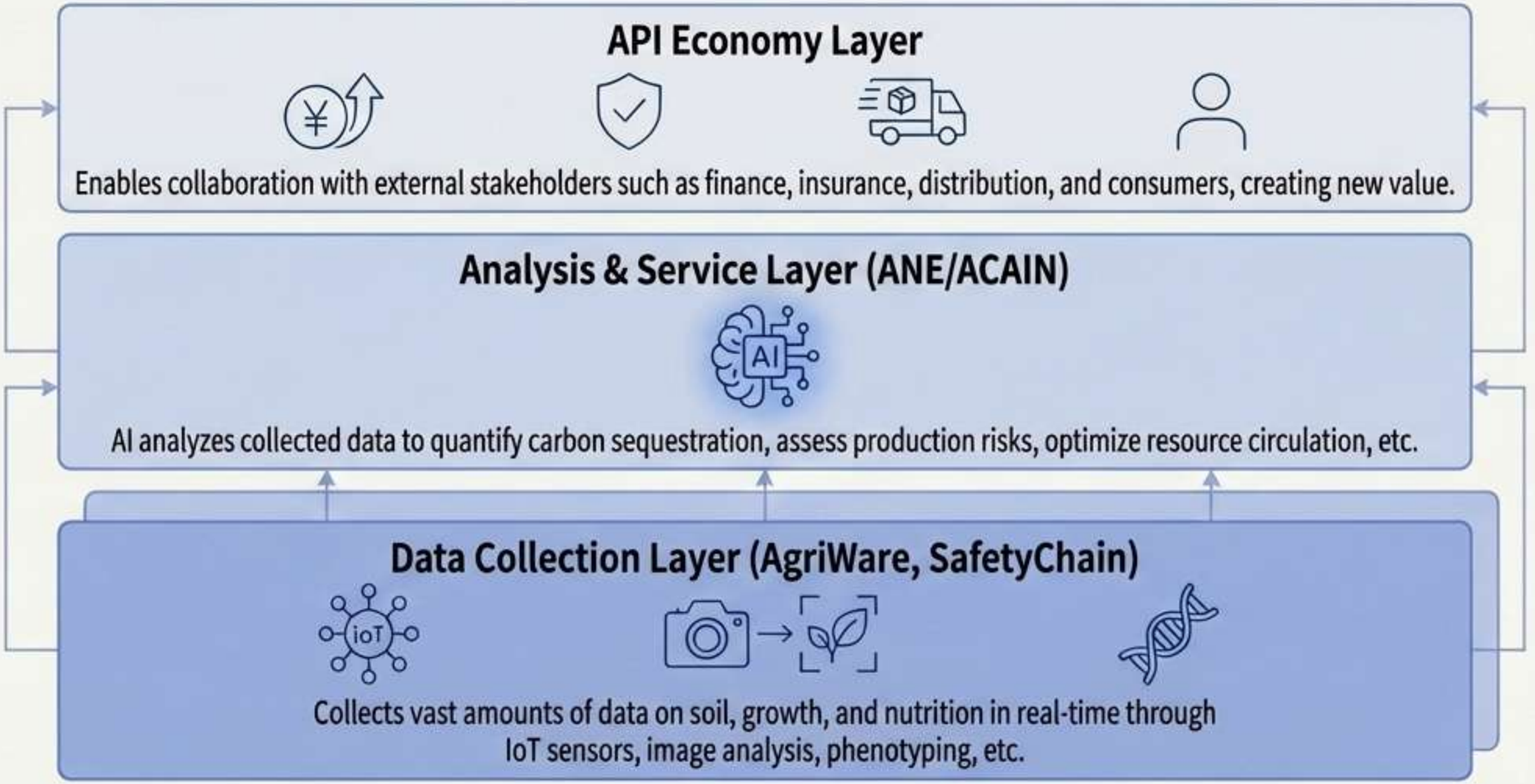


Economic Impact: Compared to traditional copper-based fungicides ($200-400/ha), costs are significantly reduced (estimated $80-150/ha), while also avoiding the risks of resistant bacteria and soil contamination.

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Foundation for Scale and Trust: Digital Platform AGRIX

AGRIX is the integrated digital infrastructure that enables the Measurement, Reporting, and Verification (MRV) of MBT55's biological impact for global deployment.

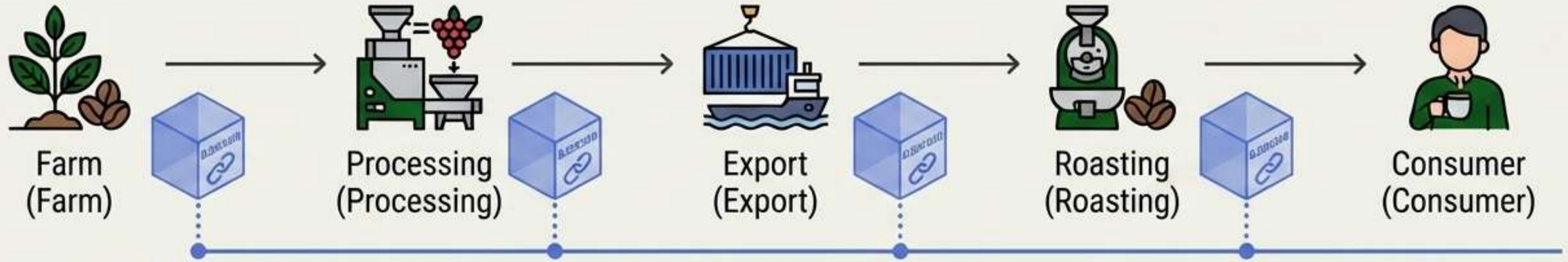


AGRIX is an edge-cloud integrated digital ecosystem that connects agriculture, bioeconomy, and climate tech.

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Case Study ② Coffee Industry: Transparency in Distribution and Value

AGRIX solves the problems of “opaque profit distribution” and “greenwashing” in the coffee supply chain.



Blockchain Traceability

Records MBT55 application history at the farm, carbon sequestration amount, and quality scores on the blockchain. Provides tamper-proof information.

Impact: Automatic settlement via smart contracts increases the profit return rate to producers from the conventional 20-30% to 60-70%.

Reliable Carbon Credit Issuance

IoT sensors and AI monitor and quantify soil carbon dynamics in real-time (VCS/Gold Standard compliant).

Impact: Based on reliable MRV, creates additional revenue of $330-$1,030 per hectare annually. Contributes to corporate Scope 3 reduction.

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Global Impact Visualization: AGRIX Dashboard

AGRIX is an operating system that tracks and manages regeneration progress globally in real-time. This dashboard comprehensively visualizes environmental, economic, and social value creation.

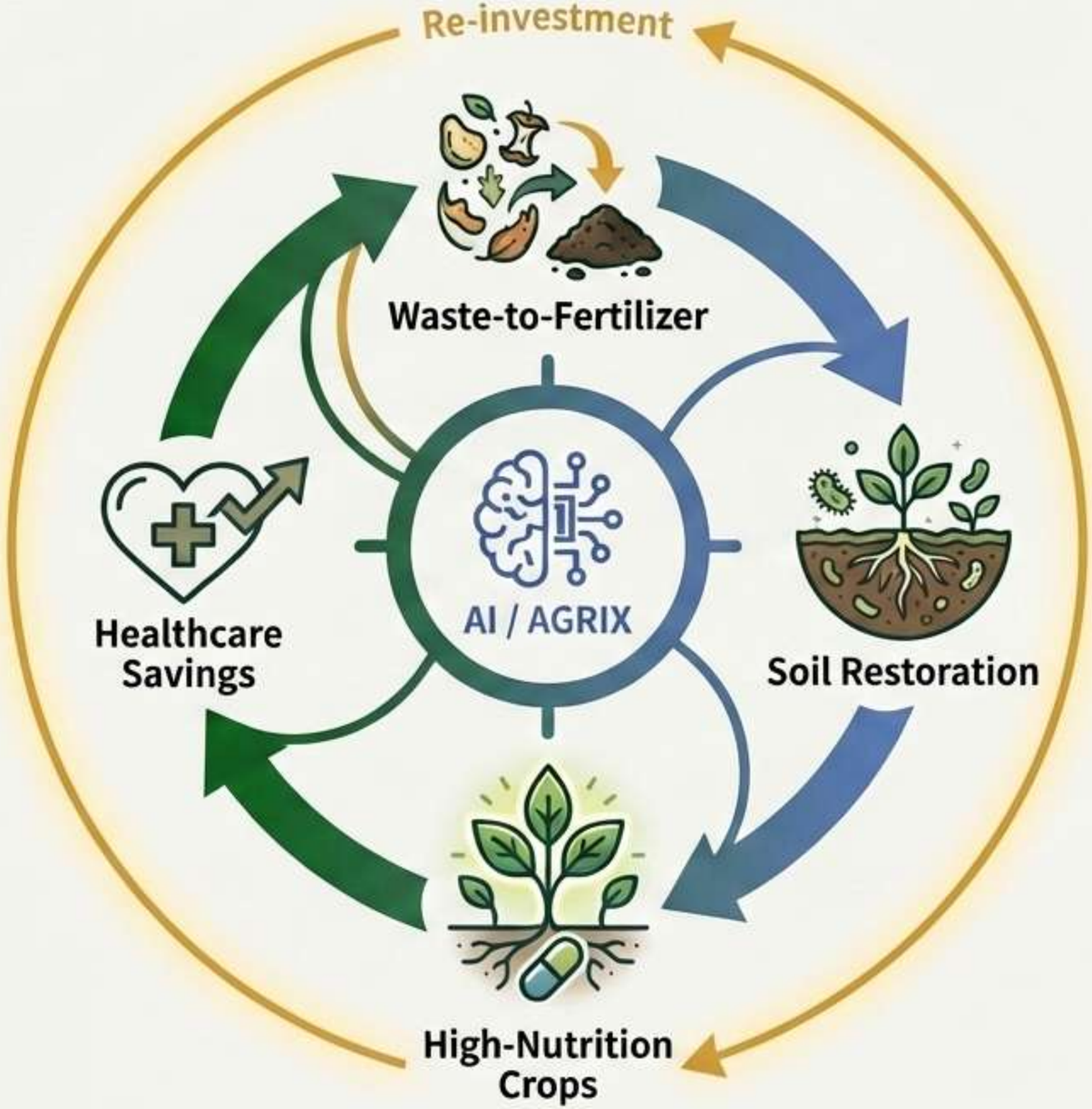


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New Economic OS: Plant-Based Planet Economy (PBPE)

PBPE is a planet-regenerative economic sphere built upon the biological engine of MBT55 and the digital trust foundation of AGRIX.

- **Philosophy of PBPE:** Construct an infrastructure where ‘good productive activities’, not speculation, are financially rewarded. If Bitcoin is a ‘store of value’, PBPE is a currency that records the ‘creation of value’.
- This is the world’s first triple-win model that simultaneously creates value in the areas of agriculture, waste management, and healthcare.



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PBPE Financial Products: A New Asset Class Investing in Earth Regeneration

PBPE consists of three core financial products that directly contribute to Earth's regeneration. These are backed by real economic results, not for speculative purposes.



1. PBPE Green Bond

- **Purpose:** Fundraising for the expansion of MBT55-introduced farms and AGRIX infrastructure development.
- **Underlying Asset:** Carbon absorption volume by MBT55 and cash flow from productivity improvement.
- **Feature:** Has composite social value of "Carbon Sequestration x Food Production x Healthcare Cost Reduction".



2. PBPE Token

- **Purpose:** A medium of value circulation within the PBPE economy.
- **Underlying Value:** Real value such as carbon absorption, agricultural productivity improvement, and food loss reduction.
- **Feature:** A structure where "production activity itself generates value".



3. PBPE Carbon Credit

- **Purpose:** Issuance of carbon absorption and emission reduction by MBT55 as credits based on international standards.
- **Underlying Asset:** CO_2 reduction volume (CO_2 reduction volume MRV'd (Measured, Reported, and Verified) by AGRIX.
- **Feature:** Ensures complete traceability with blockchain. Can be used for corporate Scope 3 reduction.

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PBPE Tokenomics: Proof of Regeneration

PBPE token is designed with a fundamentally different philosophy from Bitcoin's Proof of Work.
Its value is created solely by "activities that regenerate the Earth", not by computational power.

Proof of Work



Bitcoin: High Energy Consumption & Computational Effort

Issuance Model: Proof of Regeneration (PoR)

Positive Planetary Outcomes (Carbon Balance Improvement ΔC, Yield Improvement ΔY, Loss Reduction ΔL, ...)

→



Verified by AGRIX Platform

→



PBPE Token Minted

Proof of Regeneration



PBPE: Positive Real-World Outcomes



Value Anchor:

- **Carbon Market:** 1 PBPE is always exchangeable for a fixed amount of PBPE Carbon Credit. This allows the carbon market price to support the token's floor value.
- **Real Demand:** Can be used as a means of payment for PBPE certified products (coffee, vegetables, etc.), and real economy demand supports the value.



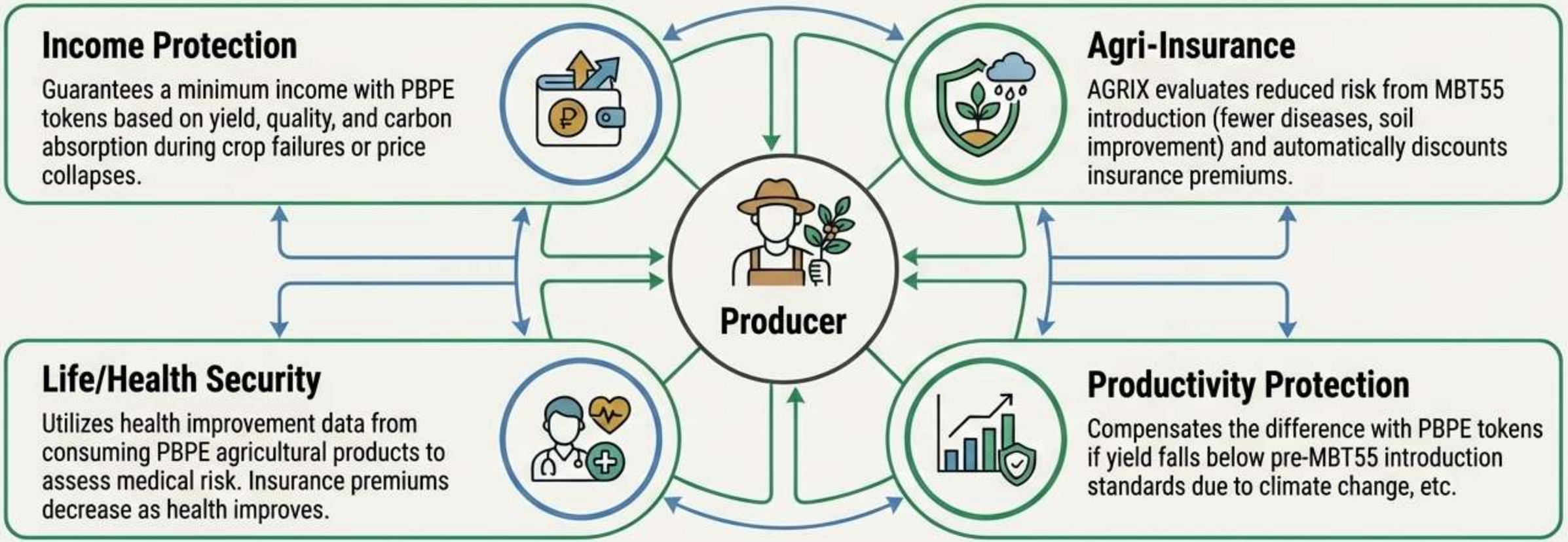
Speculation Suppression Mechanism:

- **Regeneration Fee:** A small fee is charged for short-term trading, which is automatically distributed to local farmers and research funds. It's designed so that "the more speculative the trading, the more forced social contribution occurs".

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Case Study ③ Coffee Industry: An Integrated Safety Net Supporting Producers' Lives

PBPE builds the world's first "Regenerative Safety Net" to solve challenges faced by coffee producers, such as poverty, disasters, and health insecurities. This aligns perfectly with the philosophy advocated by Starbucks founder Howard Schultz.



Impact: Producers are protected from risk, become healthier, and can form an "asset" called carbon absorption. This is an economic system for producers to break the cycle of poverty and become the protagonists of planetary regeneration.

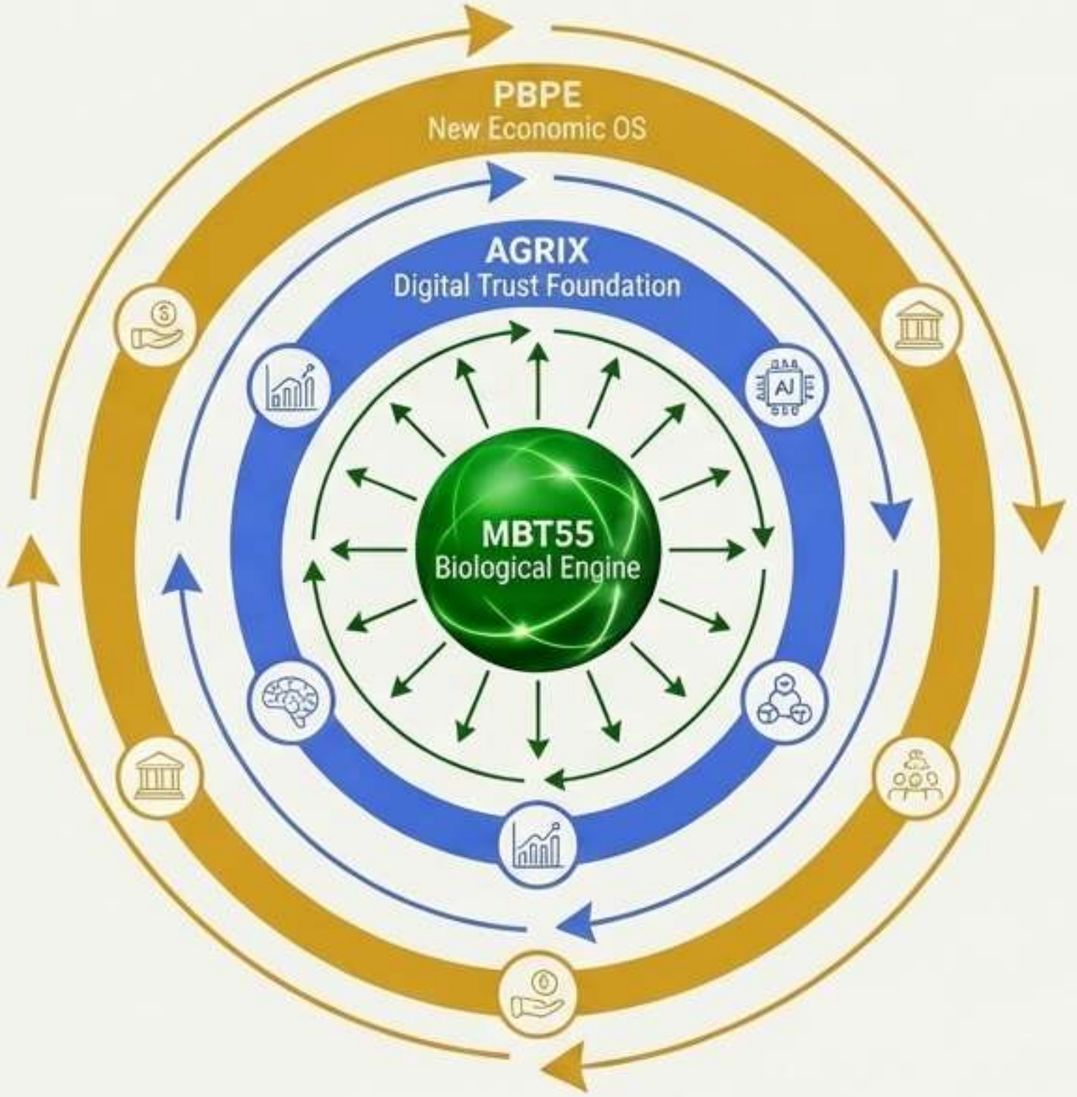
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Self-Amplifying Earth Regeneration System: Towards a New Paradigm of Circular Carbon Management

Our proposal is not merely a collection of individual technologies. It is a self-amplifying economic system that accelerates and optimizes the Earth's own regulatory functions.

- The biological engine (MBT55) transforms waste into valuable resources,
- The digital trust foundation (AGRIX) measures and verifies that value,
- The new economic OS (PBPE) financially rewards that value and encourages reinvestment.

This model goes beyond the IPCC's 'CDR (Carbon Dioxide Removal)' to provide a new framework called 'Circular Carbon Management (CCM)', which accelerates the carbon cycle while enhancing ecosystem resilience. This is a blueprint for a future where economic growth and planetary health are synonymous.



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